

EXPERIMENT – 1

Setting Up a Linux Platform Using Oracle VirtualBox and Kali Linux

Aim

To set up a Linux working environment using Oracle VirtualBox and Kali Linux Virtual Machine for programming, networking, and cybersecurity laboratory experiments.

Objective

To understand different methods of installing Linux, learn the concept of virtualization, and configure a virtual machine successfully.

Theory

Linux is a powerful open-source operating system widely used in software development and cybersecurity. A Virtual Machine (VM) allows a user to run an operating system inside another operating system without affecting the host machine. A Hypervisor is software that creates and manages virtual machines. Hypervisors are of two types: Type-1 (Bare Metal) and Type-2 (Hosted). Oracle VirtualBox is a Type-2 Hypervisor that runs on top of an existing operating system.

Requirements

Oracle VirtualBox, Kali Linux Virtual Machine Image, Host Operating System (Windows/Linux), and Internet Connection.

Procedure

1. Downloaded Oracle VirtualBox from the official website. 2. Downloaded the Kali Linux VirtualBox image from the Kali Linux website. 3. Extracted the downloaded VM files. 4. Installed Oracle VirtualBox on the host system. 5. Imported the Kali Linux Virtual Machine into VirtualBox. 6. Configured the VM settings. 7. Started the virtual machine and verified successful booting of Kali Linux.

Observations

The VirtualBox software installed successfully. The Kali Linux virtual machine was imported without errors. The VM booted correctly and the Kali Linux desktop environment loaded successfully.

Result

A fully functional Kali Linux virtual machine was successfully created and executed using Oracle VirtualBox.

Conclusion

The Linux platform was successfully set up using Oracle VirtualBox and Kali Linux. The environment is now ready for future programming and cybersecurity practical experiments.

Screenshots and Evidence

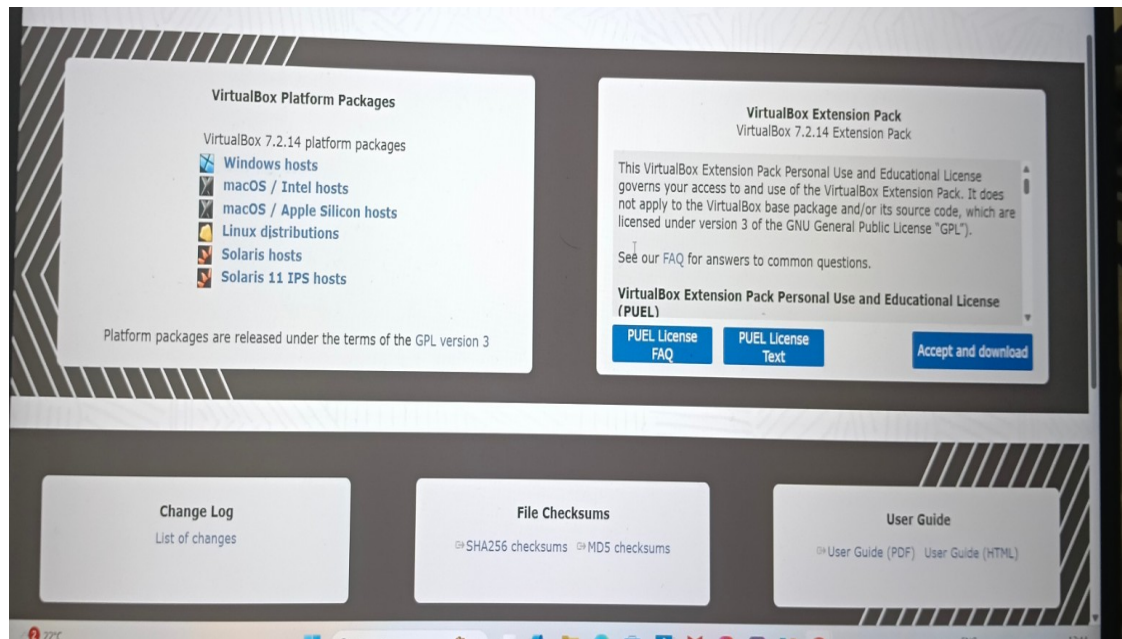


Figure 1: Oracle VirtualBox Download Page Showing Platform Packages and Extension Pack

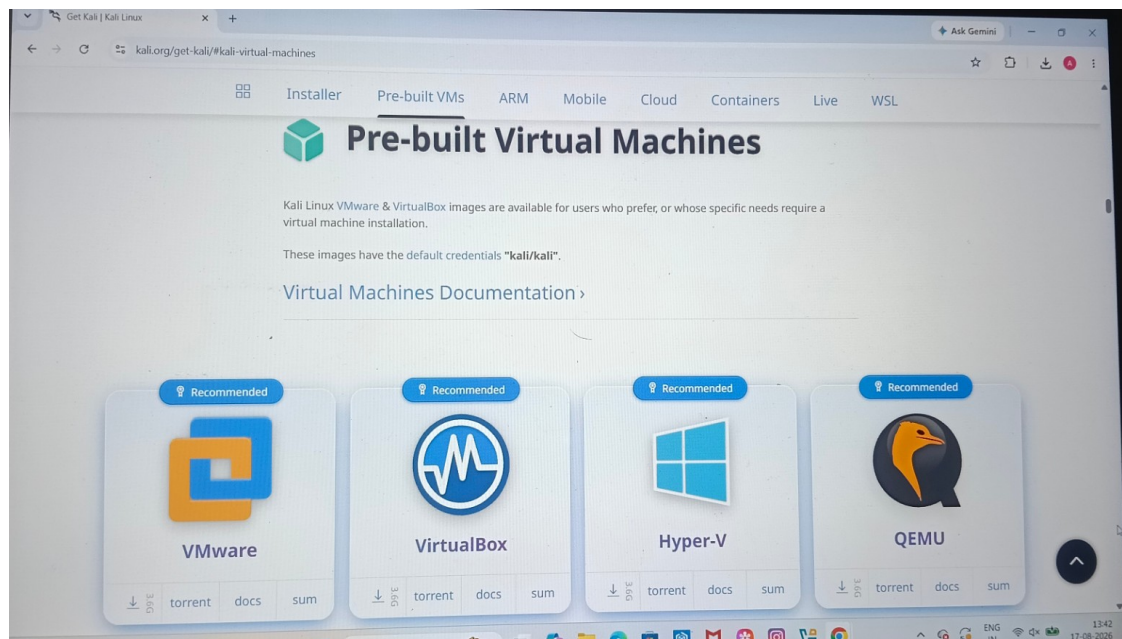


Figure 2: Kali Linux Pre-built Virtual Machine Download Page

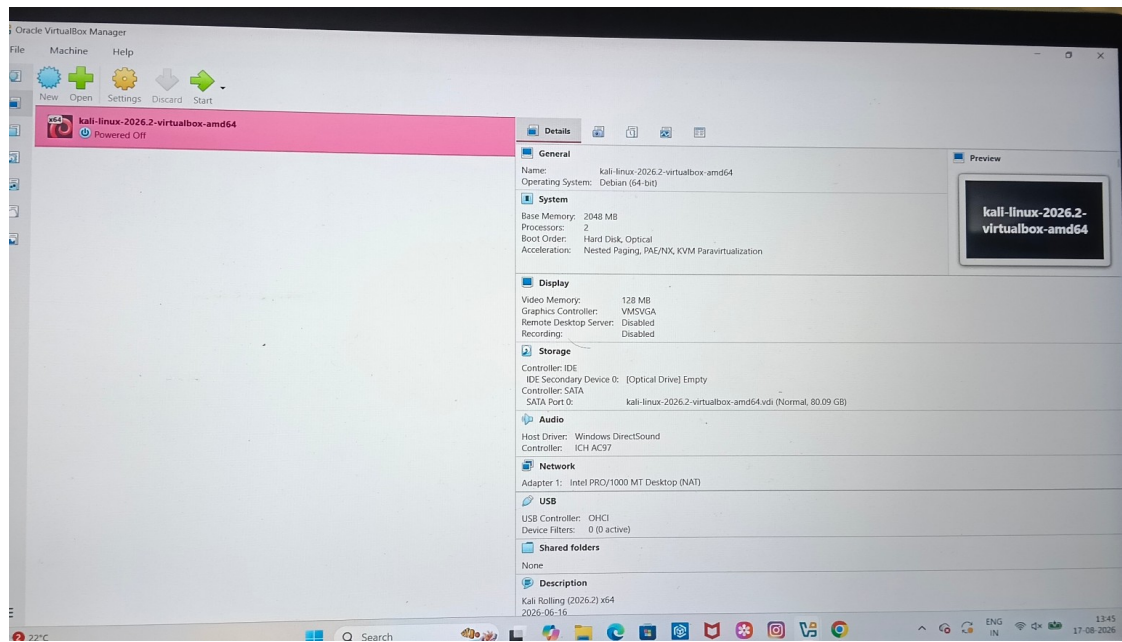


Figure 3: Oracle VirtualBox Manager Showing Imported Kali Linux Virtual Machine

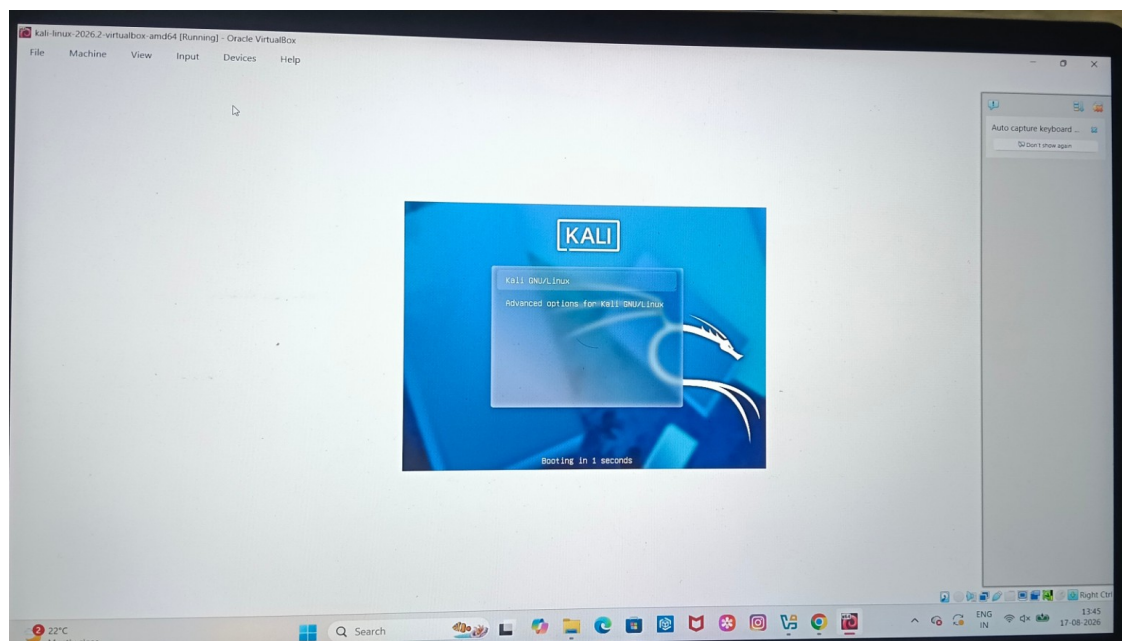


Figure 4: Kali Linux Boot Menu Running Inside Oracle VirtualBox

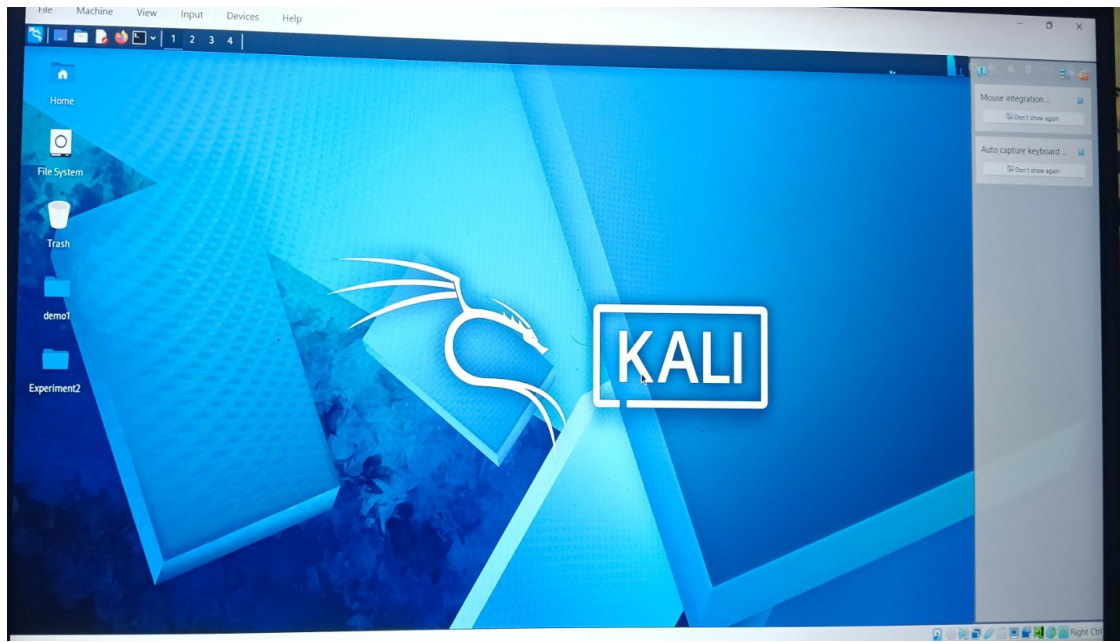


Figure 5: Kali Linux Desktop Environment Successfully Loaded